

Review article: Numerical Methods for Engineering Quantum Computers

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ABSTRACT: The main techniques in physically storing information in a “Quantum superposition state” and inducing “entanglement” amongst pairs of “Qubits” are (a) trapped ions, (b) superconducting circuits (e.g. Unimon Qubit), (c) Nitrogen-vacancy centers in diamond, and (d) Photonic Qubits.

Each of these vehicles for quantum information technology have their advantages and disadvantages. Important properties to consider are (i) resilience to noise, (ii) ability to communicate “quantum” information over long distances, (iii) decoherence time, (iv) operating temperature, (v) scalability of multi Qubit entanglement, and (vi) being amenable to strategic control. Right now, Quantum computers are in the “Noisy Intermediate-Scale Quantum” (NISQ) era. In an effort to constantly improve the design of quantum computers, scientists and engineers answer questions regarding the aforementioned Qubit technology either experimentally or computationally. In this review article, it is the computational methods which are reported on. It is the objective of this review article to serve as a reference on the subject matter of “Numerical Methods for Engineering Quantum Computers” for use by students and established researchers. As a side benefit, it is expected that understanding the mathematical models and computational algorithms used to engineer Quantum Computers will also lead to improved algorithms for utilizing Quantum Computers.

As quoted from Shalf[21], “Moore’s Law [1] is a techno-economic model that has enabled the IT industry to double the performance and functionality of digital electronics roughly every 2 years within a fixed cost, power and area. This expectation has led to a relatively stable ecosystem (e.g. electronic design automation tools, compilers, simulators and emulators) built around general-purpose processor technologies, such as the $\times 86$, ARM and Power instruction set architectures. However, within a decade, the technological underpinnings for the process that Gordon Moore described will come to an end, as lithography gets down to atomic scale. At that point, it will be feasible to create lithographically produced devices with dimensions nearing atomic scale, where a dozen or fewer silicon atoms are present across critical device features, and will therefore represent a practical limit for implementing logic gates for digital computing [2]. Indeed, the ITRS (International Technology Roadmap for Semiconductors), which has tracked the historical improvements over the past 30 years, has projected no improvements beyond 2021, as shown in figure 1, and subsequently disbanded, having no further purpose. The classical technological driver that has underpinned Moore’s Law for the past 50 years is failing [3] and is anticipated to flatten by 2025, as shown in figure 2. Evolving technology in the absence of Moore’s Law will require an investment now in computer architecture and the basic sciences (including materials science), to study candidate replacement materials and alternative device physics to foster continued technology scaling.”

References one through three in the above quote correspond to the following references respectively: [18][16][17].

In Figure 3 of the article by Shalf[21], a roadmap is provided for possible paths forward when the density of circuits on classical computers exceeds a critical value. There are three categories: (a) roadmap for the next ten years, (b) 20 years, and (c) “Decades beyond exascale,” “New Models of Computation.” For category (c), some of the prospective technology listed is: (i) approximate computing, (ii) adiabatic reversible, (iii) Analog, (iv) Neuromorphic, and (v) quantum.

Of the “new models of computation” listed in Shalf’s article[21], this review article will address the “Numerical Methods for Engineering Quantum Computers” aspects associated with the emerging “quantum computing” paradigm. As outlined by Gamble[7], the development of reliable quantum computers will have a transformative effect on computer technology. That being said, right now we are in the “Noisy Intermediate-Scale Quantum” [4] (NISQ) era. The purpose of this review article is to bring to the forefront the current research being done, from the computational perspective, in order to move beyond the NISQ era.

1 preliminaries

In order to appreciate the complexities associated with designing a quantum computer, it is suggested that one first gain some intuition on electricity and magnetism.

Classical electromagnetism is a very complex topic, in of itself. One could choose as a starting point Maxwell’s equations which are,

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0} \quad (\text{Gauss’s Law for Electricity}) \quad (1)$$

$$\nabla \cdot \mathbf{B} = 0 \quad (\text{Gauss’s Law for Magnetism}) \quad (2)$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} \quad (\text{Faraday’s Law of Induction}) \quad (3)$$

$$\nabla \times \mathbf{B} = \mu_0 \left(\mathbf{J} + \epsilon_0 \frac{\partial \mathbf{E}}{\partial t} \right) \quad (\text{Ampere-Maxwell’s Law}) \quad (4)$$

Also, there is the Lorentz force law:

$$\mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B})$$

A good textbook that describes these fundamentals is by Purcell and Morin third edition[19].

Maxwell’s equations, and relativity though are not sufficient to describe the quantum and particle nature of matter and light. Matter and Light, when considering small enough scales, have a dual particle and wave property to them. Also Matter and light possess discrete “quantum,” energy levels. For a good reference, see the following quantum mechanics textbook[20]. Experiments which cannot be explained by classical electromagnetism are the Stern-Gerlach experiment[8], the photo-electric effect[22], and the electron double slit experiment[1], amongst many more.

The mathematical model that describes the dynamics of matter and light, including “quantum” properties and the dual particle wave property is Schrodinger’s equation:

$$i\hbar \frac{\partial}{\partial t} \Psi(\mathbf{r}, t) = \hat{H} \Psi(\mathbf{r}, t)$$

where: i is the imaginary unit $\hbar$ is the reduced Planck constant $\frac{\partial}{\partial t}$ is the partial derivative with respect to time $\Psi(\mathbf{r}, t)$ is the wave function $\hat{H}$ is the Hamiltonian

operator $\mathbf{r}$ is the position vector t is time The time-independent Schrödinger equation can be written as:

$$\hat{H}\psi(\mathbf{r}) = E\psi(\mathbf{r})$$

where:

$$\hat{H}$$

is the Hamiltonian operator

$$\psi(\mathbf{r})$$

is the time-independent wave function

$$E$$

is the energy of the system. When discussing the engineering of quantum computers, relativistic effects do not have to be considered. Also, nonlinear Shrodinger mathematical models do not have to be considered.

For the applications of quantum computing and quantum information processing, it is argued that relativity effects can be ignored since the rest energy possessed by an electron is $E = m_e c^2$ which is significantly larger than the energy required for inducing a superposition state or the energy to carry out a unitary operator on a current quantum state.

“Quantum computers operate on energy levels far below those where relativistic effects become noticeable. The speeds of qubits and the interactions between them are slow enough that the effects of time dilation and length contraction are practically insignificant. ”

2 Quantum Algorithms for benchmarking Quantum Computers

<https://www.ibm.com/quantum/qiskit>

- Gopalakrishnan Meena et al[9]
- Jin et al[12]
- Jin et al (heat equation)[13]
- Beck et al[2]
- Harrow et al[11]
- Daribayev et al[5]
- Bravo-Prieto et al[3]

3 Quantum Algorithms with error correction logic

- Kubischta, Eric and Teixeira, Ian[14]

4 Control and optimization of numerical models for simulating quantum computers

- Günther et al[10]
- Fei et al[6]

4.1 Control of open quantum systems

The Lindblad master equation for Open quantum systems[15] is

$$\dot{\rho}(t) = -i[H(t), \rho(t)] + \ell(\rho(t)) \quad 0 \leq t \leq T \quad (5)$$

$$\ell(\rho) = \sum_{q=1}^Q \sum_{l=1}^2 \ell_{lq} \rho \ell_{lq}^\dagger - \frac{1}{2} (\ell_{lq}^\dagger \ell_{lq} \rho + \rho \ell_{lq}^\dagger \ell_{lq}) \quad (6)$$

$$N = \Pi_{q=1}^Q n_q \quad (7)$$

$$\rho(t) \in \mathbb{C}^{N \times N} \quad (8)$$

From [15], given the pure state ensemble,

$$\{q_i, |\psi\rangle_i\}_{i=1}^N, \quad (9)$$

the density operator is

$$\rho = \sum_i q_i |\psi\rangle_i \langle\psi|_i. \quad (10)$$

$$H = H_{sys} + H_c(t) \quad (11)$$

The control Hamiltonian has the form

$$H_c(t) = \sum_{q=1}^Q d_q(\alpha_q, t) a_q + d_q(\alpha_q, t)^*(t) a_q^\dagger. \quad (12)$$

The objective function has the form

$$\mathcal{G}(\alpha) = \frac{1}{M} \sum_{i=1}^M \left(\beta_i J(\rho_i^{target}, \rho_i(T)) + \gamma_1 \int_0^T \beta(t) J(\rho_i^{target}, \rho_i(t)) dt \right) + \gamma_2 \|\alpha\|_2^2 / 2 \quad (13)$$

In Quandary the following choices for J are[10]

$$J_{Frob}(\rho^{target}, \rho) = \|\rho^{target} - \rho\|_F^2/2 \quad (14)$$

$$J_{Trace}(\rho^{target}, \rho) = 1 - Tr((\rho^{target})^\dagger \rho) \quad (15)$$

$$J_{N_m}(\rho) = Tr(N_m \rho) \quad (16)$$

4.2 Switching time optimization Control of binary quantum systems

$$\dot{X}(t) = -iH(t)X(t) \quad 0 \leq t \leq T \quad (17)$$

$$H(t) = H^{(0)} + \sum_{j=1}^J u_j(t)H^{(j)} \quad 0 \leq t \leq T \quad (18)$$

$$\sum_{j=1}^J u_j(t) = 1 \quad 0 \leq t \leq T \quad (19)$$

$$u_j(t) \in \{0, 1\} \quad j = 1, \dots, J \quad (20)$$

Cost function example ($Q = 2$):

$$\mathcal{G}(X[u_1(t), \dots, u_J(t)](T)) = 1 - \frac{1}{4} |\text{Trace} \{ X_{CNOT}^\dagger X(T) \}| \quad (21)$$

- compare VQE on classical computer vs IBM hardware to demonstrate the need for improved hardware.
- compare QUANDARY to other quantum optimal control algorithms. inverse problems.
- compare Error correction algorithms.
- quantum and hybrid classical/quantum algorithms for benchmarking quantum computers.

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